

Using the lemma if  $\int_{t_0}^{t_f} g(t) \delta x(t) dt = 0$  then

$g(t) = 0$  at every point over the interval  $[t_0, t_f]$

so our necessary condition for optimality is (as  $\delta x(t)$  and  $\delta u(t)$  are not equal to zero)

$$\left[ \frac{\partial \Lambda(.)}{\partial x(t)} - \frac{d}{dt} \left( \frac{\partial \Lambda(.)}{\partial x'(t)} \right) \right]_* = 0 \quad \dots (23.14)$$

In case  $\mathbf{x}$  is  $n \times 1$  vector

$$\text{then} \left[ \frac{\partial \Lambda(.)}{\partial x(t)} - \frac{d}{dt} \left( \frac{\partial \Lambda(.)}{\partial x'(t)} \right) \right]_* = O_{n \times 1} \quad \dots (23.15)$$

In order to solve these differential equations we need boundary condition.

$$\text{Similarly, } \frac{\partial \Lambda(.)}{\partial u(t)} = 0 \quad \dots (23.16)$$

if  $\mathbf{u}(t)$  is scalar

$$\text{and } \frac{\partial \Lambda(.)}{\partial u(t)} = O_{m \times 1} \quad \dots (23.17)$$

if  $\mathbf{u}$  is  $m \times 1$  vector

Because 1<sup>st</sup> two terms are zeros in equation 23.13

$$\partial_J \approx \Lambda(.)|_{t=t_f} \cdot \delta t_f + \left( \frac{\partial \Lambda(.)}{\partial x'(t)} \right) \delta x(t)|_{t=t_f} \quad \dots (23.18)$$

After some algebraic calculation which takes into consideration  $\delta x'(t)$ .  $\delta t_f \approx 0$

$$\text{We get } \partial_J \approx \left[ L(.) - \frac{\partial \Lambda(.)}{\partial x'(t)} x'(t) \right]_{t=t_f} \delta t_f + \left. \frac{\partial \Lambda(.)}{\partial x'(t)} \right|_{t=t_f} \cdot \delta x_f = 0 \quad \dots (23.19)$$

Therefore in order to make necessary condition the equation 23.19 needs to be solved. This is general boundary condition this takes into account that  $x_f$  and  $t_f$  are both free. We are getting two boundary condition as there are two terms in equation (23.19). (Note that we need to make  $\partial_J$  as small as possible so we set 23.19 to zero)

If  $\mathbf{x}$  and  $\mathbf{u}$  are vector of dimension  $n \times 1$  and  $m \times 1$  respectively.

then the boundary condition would be

$$\left[ \Lambda(.) - \left( \frac{\partial \Lambda(.)}{\partial x'(t)} \right)^T x'(t) \right]_{t=t_f} \delta t_f + \left. \left( \frac{\partial \Lambda(.)}{\partial x'(t)} \right)^T \right|_{t=t_f} \cdot \delta x_f = 0 \quad \dots (23.20)$$

In summary we need to solve

equations (23.14) (23.16) (23.19) when  $x$  and  $u$  are scalars and equations (23.17) (23.15) (23.20) when  $x$  and  $\mathbf{u}$  are vectors. Solving these pairs of equation will give us  $\mathbf{u}^*(t)$  which in turn gives us  $\mathbf{x}^*(t)$ .

## 23.3 HAMILTONIAN FORMATION OF OPTIMAL CONTROL PROBLEM

Let us restate what we have learned so far

$$\mathbf{x}'(t) = \mathbf{f}(\mathbf{x}(t), \mathbf{u}(t), t)$$

Given the above constraint we have to optimise

$$J(.) = S(\mathbf{x}(t_f), t_f) - S(\mathbf{x}(t_0), t_0) + \int_{t_0}^{t_f} v(\mathbf{x}(t), \mathbf{u}(t), t) dt$$

We have derived necessary condition

$$\left[ \frac{\partial \Lambda(.)}{\partial \mathbf{x}(t)} - \frac{d}{dt} \left( \frac{\partial \Lambda(.)}{\partial \mathbf{x}'(t)} \right) \right] = 0$$

and boundary condition

$$\left[ \Lambda(.) - \left( \frac{\partial \Lambda(.)}{\partial \mathbf{x}'(t)} \right) \mathbf{x}'(t) \right]_{*t=t_f} \delta t_f + \left. \frac{\partial \Lambda(.)}{\partial \mathbf{x}'(t)} \right|_{*t=t_f} \delta \mathbf{x}_f$$

where \* stands for optimal value of  $\mathbf{u}^*(t)$  which in turn gives us  $\mathbf{x}^*(t)$

$$\text{and we know } \Lambda(.) = H(\mathbf{x}(t), \mathbf{u}(t), \lambda(t), t) + \left( \frac{\partial S(.)}{\partial \mathbf{x}(t)} \right) \mathbf{x}'(t) + \frac{\partial S(.)}{\partial t} - \lambda(t) \mathbf{x}'(t)$$

**Note:**

$$\begin{aligned} H &= \text{Hamiltonian function (this is free from } \mathbf{x}'(t) \text{)} \\ &= V(\mathbf{x}(t), \mathbf{u}(t), t) + \lambda(\mathbf{x}(t), \mathbf{u}(t), t) \end{aligned}$$

Using equation 23.10 in equation (23.14) i.e., the value of  $L(.)$  is used in necessary condition.

$$\text{Using the fact that } \frac{\partial H}{\partial \mathbf{x}'(t)} = 0 \text{ and } \frac{\partial}{\partial \mathbf{x}'(t)} \left[ \frac{\partial S}{\partial t} \right] = 0$$

and after simplification we get

$$\frac{\partial H(.)}{\partial \mathbf{u}(t)} = 0 \quad \dots (23.21) \text{ for } \mathbf{u}(t) \text{ as scalar}$$

$$\left( \frac{\partial H(.)}{\partial \mathbf{u}(t)} \right) = \mathbf{0}_{m \times 1} \quad \dots (23.22) \text{ for } \mathbf{u}(t) \text{ as } m \times 1 \text{ vector}$$

$$\left( \frac{\partial H(.)}{\partial \mathbf{x}(t)} \right) = -\lambda'(t) \quad \dots (23.23) \text{ for } \mathbf{x}(t) \text{ is scalar}$$

$$\left( \frac{\partial H(.)}{\partial \mathbf{x}(t)} \right)_{n \times 1} = -\lambda'(t) \quad \dots (23.24) \text{ for } \mathbf{x}(t) \text{ is } n \times 1 \text{ vector}$$

$$\left( \frac{\partial H(.)}{\partial \lambda(t)} \right) = \mathbf{x}'(t) \quad \dots (23.25) \text{ for } \mathbf{x}(t), \mathbf{u}(t) \text{ as scalar}$$

$$\left( \frac{\partial H(.)}{\partial \lambda(t)} \right) = \mathbf{x}'(t) \quad \dots (23.26) \text{ for } \mathbf{x} \text{ is } (n \times 1) \text{ vector.}$$

Equation 23.25 or 23.26 is called state equation

Equation 23.23 or 23.24 is called co-state equation

For multivariate case  $\lambda$  is  $(n \times 1)$  vector as  $\mathbf{x}$  is  $n \times 1$  vector

Solving state and co-state equation and  $\frac{\partial H(.)}{\partial \mathbf{u}(t)} = 0$  will give us  $\mathbf{u}^*(t)$  and in turn  $\mathbf{x}^*(t)$

In order to get boundary condition under Hamiltonian formulation, we need to put (23.10) in equation (23.20). after simplification, we get the boundary condition.

$$\left[ H(.) + \frac{\partial S(.)}{\partial t} \right]_* \Big|_{t=t_f} \delta t_f + \left( \frac{\partial S(.)}{\partial \mathbf{x}(t)} - \lambda(t) \right)_* \Big|_{t=t_f} \delta \mathbf{x}_f = 0 \quad \dots (23.27)$$

The above equation is for  $x^*(t)$  and  $\mathbf{u}^*(t)$  as scalar. When  $\mathbf{x}^*(t)$  and  $\mathbf{u}^*(t)$  are vectors of  $(n \times 1)$  and  $(m \times 1)$  dimension then

$$\left[ H(.) + \frac{\partial S(.)}{\partial t} \right]_* \Big|_{t=t_f} \delta t_f + \left( \frac{\partial S(.)}{\partial \mathbf{x}(t)} - \lambda(t) \right)_* \Big|_{t=t_f}^T \delta \mathbf{x}_f = 0 \quad \dots (23.28)$$

Is the boundary condition equation for multivariate case.

## Check Your Progress 1

1) Optimize  $V = \int_0^1 -u^2 dt$  s.t.  $y' = y + u$  and  $y(0) = 1, y(1) = 0$

.....  
.....

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## 23.4 HAMILTONIAN OPTIMALITY CONDITION IN SPECIAL CASES

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**Case A:** Fixed final time and fixed final state.

For solving this type of problems there is no boundary condition we need to solve only (23.21)/(23.22), (23.23)/(23.24), (23.25)/(23.26) to get optimality condition

**Case B:** Free final time and fixed final state here  $\delta t_f$  is free and  $\delta \mathbf{x}_f = 0$

Hence for optimality, besides solving the three equations we need to solve one more boundary condition needs to be solved.

i.e.,  $H(.) + \frac{\partial S(.)}{\partial t} = 0 \quad \dots (23.29)$

Case C: Fixed final time and free final state.

$\Rightarrow \delta t_f = 0$  and  $\delta x_f$  is free.

Hence for optimality we need to solve the 3 set of equations as in case A and

$$\left[ \frac{\partial S}{\partial x} - \lambda \right]_{t=t_f} = 0 \quad \dots(23.30)$$

(for univariate case)

$$\text{for multivariate case } \left[ \frac{\partial S}{\partial x} - \lambda(t) \right]^T = O_{n \times 1} \quad \dots (23.31)$$

**Case D:** Free final time and free final state.

For optimally besides solving 3 sets of equation we need to solve two boundary condition.

$$H(.) + \frac{\partial S(.)}{\partial t} = 0 \quad \dots(23.32)$$

$$\left[ \frac{\partial S}{\partial x} - \lambda(t) \right] = 0 \quad \dots (23.33) \text{ (For univariate } x) \text{ or}$$

$$\left[ \frac{\partial S}{\partial x} - \lambda(t) \right]^T = O_{n \times 1} \quad \dots (23.34) \text{ (For multivariate 'x' case)}$$

## 23.5 TRANSVERSALITY CONDITION

Sometimes, the terminal value of the state variable  $x(t_f) = x_f$  is restricted in sign. This is usually the case in economic application of the maximum principle, where a typical state variable is the stock of productive capital in the economy. We have as transversality condition.

$$x_f \geq 0, x_f \lambda_f = 0 \quad \dots (23.35)$$

which implies that either  $x_f = 0$  or else  $\lambda_f = 0$ . If on the contrary,  $x_{tf}$  is not restricted in value or sign, then we must have as transversality condition  $x_{tf} = 0$

if the planning problem has an infinite time horizon, the transversality condition becomes

$$\lim_{t_f \rightarrow \infty} x_{tf} \geq 0, \lim_{t_f \rightarrow \infty} x_{tf} \lambda_{tf} = 0 \quad \dots (23.36)$$

when  $x_{tf}$  is restricted in sign and  $\lim_{t_f \rightarrow \infty} \lambda_{tf} = 0$

when  $x_{tf}$  is not restricted in sign or value.

## 23.6 THE DISCOUNTED PROBLEM

In case of especially economic growth problems the global intertemporal objective is the result of aggregative overtime an instantaneous objective function subject to sometime discount  $\theta$ . That is function

$f(x_t, u_t, t)$  is of the form

$$f(x_t, u_t, t) = e^{-\theta t} g(x_t, u_t, t)$$

Where the time discount makes the net contribution of a given level of the objective function  $g(\cdot)$  to be lower the farther it occurs into the future.

Therefore, our problem is

$$\text{Max}_{u_t} \int_0^T e^{-\theta t} g(x_t, u_t, t)$$

Subject to constraint  $x'(t) = h(x_t, u_t, t)$  and  $x_{t0} = x_0$  given

As discussed earlier we can write the Hamiltonian as

$$\begin{aligned} H(\cdot) &= e^{-\theta t} g(\cdot) + \lambda_t h(\cdot) \\ &= e^{-\theta t} [g(\cdot) + \lambda_t e^{\theta t} h(\cdot)] \end{aligned} \quad \dots (23.37)$$

We define the current value multiplier  $\mu^t = \lambda_t e^{\theta t}$  which provides us with the shadow price or marginal value of the state variable  $x_t$  at time  $t$ .

We define the current value of Hamiltonian  $H^*$

$$H^*(\cdot) = e^{\theta t} H(\cdot) = g(\cdot) + \mu_t h(\cdot) \quad \dots (23.38)$$

Where it is easy to see that the choice of  $u_t$  maximizing the current value Hamiltonian  $H^*$  is same as maximizing the Hamiltonian  $H(\cdot)$  (ref equation 23.37)

Using the same optimality condition as discussed earlier

$$\frac{\partial H(\cdot)}{\partial u_t} = 0 \Rightarrow \frac{\partial e^{-\theta t} H^*(\cdot)}{\partial u_t} = e^{-\theta t} \frac{\partial H^*(\cdot)}{\partial u_t} \quad \dots (23.39)$$

(1) Form which we get

$$\frac{\partial H(\cdot)}{\partial u_t} = 0 \Leftrightarrow \frac{\partial H^*}{\partial u_t} = 0 \quad \dots (23.39a)$$

(2) We know

$$\mu_t = \lambda_t e^{\theta t}$$

$$\Rightarrow \mu'_t = \theta \mu_t + e^{\theta t} \lambda'_t$$

$$= \theta \mu_t + e^{\theta t} - \frac{\partial H(\cdot)}{\partial x_t}$$

$$= \theta \mu_t + e^{\theta t} \left[ -e^{-\theta t} \frac{\partial g(t)}{\partial x_t} - \lambda_t \frac{\partial h(\cdot)}{\partial x_t} \right]$$

$$= \theta \mu_t - \frac{\partial g(\cdot)}{\partial x_t} - \lambda_t e^{\theta t} \frac{\partial h(\cdot)}{\partial x_t}$$

$$= \theta \mu_t - \frac{\partial g(.)}{\partial x t} - \mu_t \frac{\partial h}{\partial x t} \quad \dots (23.40)$$

Simplifying equation (23.39a)

$$\frac{\partial H^*}{\partial u_t} = 0 \Rightarrow \frac{\partial g(.)}{\partial u_t} + \mu_t \frac{\partial h(.)}{\partial u_t} = 0 \quad \dots (23.41)$$

We can perform a similar transformation in the transversality conditions, to have for the infinite horizon case, in terms of the current-value multiplier,

$$\lim_{T \rightarrow \infty} x_T \geq 0 \quad \lim_{T \rightarrow \infty} e^{-\theta T} x_T \mu_T = 0 \quad \dots (23.42)$$

when  $x_T$  is restricted is sign and  $\lim_{T \rightarrow \infty} e^{-\theta T} \mu_T = 0$

when  $x_T$  is not restricted in sign or value

## 23.7 SECOND ORDER CONDITION

In order to determine the nature of optimisation (to determine the functional value as the one out of minimum or maximum) we need 2<sup>nd</sup> order condition.

For sufficient condition, we calculate

$\delta^2 J = 2^{\text{nd}}$  variation of functional

(Note we minimize 1<sup>st</sup> variation of functional to get necessary condition) \* is optimal path/trajectory).

$$\begin{aligned} \delta^2 J = \int_{t_0}^{t_f} & \left[ \delta x(t) \left[ \frac{\partial^2 H(.)}{\partial x^2(t)} \right]_* \delta x(t) + 2 \delta x(t) \left[ \frac{\partial^2 H(.)}{\partial x(t) \partial u(t)} \right]_* \delta u(t) \right. \\ & \left. + \delta u(t) \left( \frac{\partial^2 H(.)}{\partial u^2(t)} \right)_* \delta u(t) \right] dt \end{aligned}$$

We know from our earlier discussion that is  $\delta^2 J > 0$  then the functional value is minimum and if  $\delta^2 J < 0$  then the functional value is maximum.

One can write this in matrix and vector form (in quadratic form)

$$\delta^2 J = \int_{t_0}^{t_f} [\delta x(t) : \delta u(t)] \begin{bmatrix} \frac{\partial^2 H(.)}{\partial x^2(t)} & \frac{\partial^2 H(.)}{\partial x(t) \partial u(t)} \\ \frac{\partial^2 H(.)}{\partial x(t) \partial u(t)} & \frac{\partial^2 H(.)}{\partial u^2(t)} \end{bmatrix}_* \begin{bmatrix} \delta x(t) \\ \delta u(t) \end{bmatrix} dt \quad \dots (23.43)$$

Let the matrix in above equation be A

then A is a symmetric hessian matrix

when  $\mathbf{x}(t)$  is  $(n \times 1)$  vector and  $\mathbf{u}(t)$  is  $(m \times 1)$  vector then the dimension of  $A$  will be  $[(n + m) \times (n + m)]$  so what we need to do is following

- i) Solve the Hamiltonian equations to get  $\mathbf{u}^*(t)$  and  $\mathbf{x}^*(t)$
- ii) Formulate hessian matrix  $A$
- iii) Compute hessian matrix  $A$  at  $\mathbf{u}^*$  and  $\mathbf{x}^*$  to get  $A^*$
- iv) If  $A^*$  is positive definite then we can conclude that  $J^*$  is minimum or else if the matrix  $A^*$  is negative definite then  $J^*$  is maximum.

## Check Your Progress 2

1) Consider the following simple problem

$$\min \{g_0(u)\} = \int_0^1 \{(x(t))^2 + (u(t))^2\} dt$$

$$\text{Subject to } \frac{dx}{dt} = u(t), x(0) = 1$$

.....  
 .....  
 .....

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## 23.8 LET US SUM UP

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By using the similar arguments that we used to prove the theorems in unit-22 we formulate Lagrangian for continuous dynamic optimisation problems, each with equality constraint and with fixed final state. From the Lagrangian we derive the Hamiltonian functional and the state, the necessary condition for optimisation. Four types of boundary value conditions are derived. The transversality condition, the discounted problems are discussed separately as they hold considerable importance in economics. Further we state sufficient conditions that finally decide one out of maximum or minimum after necessary condition.

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## 23.9 KEYWORDS

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|                           |   |                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Boundary Condition</b> | : | The conditions that need to be solved in order to find optimal control. Boundary conditions arise because either final state or final time or both are free (unspecified) in objective functional.                                                                                                                                                                                           |
| <b>Control Variable</b>   | : | In optimal control theory a control is a variable chosen by the controller or agent to manipulate state variables similar to an actual control value. Unlike the state variable, it does not have a predetermined equation of motion. The goal of optimal control theory is to find some sequence of controls (within an admissible set) to achieve an optimal path for the state variables. |

|                                                          |   |                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hamiltonian Function</b>                              | : | A function which is formulated by taking objective functional and constraints to optimise the objective functional.                                                                                                                                                                                 |
| <b>Necessary Condition</b>                               | : | The series of equation that needs to be solved in order to find $u^*$ (optimal control variable).                                                                                                                                                                                                   |
| <b>State Variable</b>                                    | : | A state variable is one of the set of variables that are used to describe the mathematical state of a dynamic system. In our case these are series of first order differential equation.                                                                                                            |
| <b>Sufficient Condition for optimality (Hamiltonian)</b> | : | The status of hessian matrix borne out of Hamiltonian functional gives sufficient condition (or 2 <sup>nd</sup> order condition). For minimality the hessian matrix is positive definite and for maximality the hessian matrix is negative definite.                                                |
| <b>Transversality Condition</b>                          | : | In optimal control, a transversality condition is a boundary condition for terminal values of the costate variables ( $\lambda(t)$ ). It is one of the necessary conditions for optimality in case of infinite horizon optimal control problems without an end point constraint on state variables. |

## 23.10      ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

### Check Your Progress 1

1) First and foremost, since  $x_{t0}$  and  $x_{tf}$  are both fixed no boundary value condition is needed. Rewriting the problem

$$\underset{u(t)}{\text{optimize}} = \int_0^1 -u^2 dt \quad \text{s.t. } y' = y + u \text{ given the boundary value of state variable}$$

$$y(0) = 1 \quad y(1) = 0. \text{ Forming the Hamiltonian we get } H(.) = -u^2 + \lambda(y + u)$$

For the necessary condition we need to set  $\frac{\partial H(.)}{\partial u} = 0$  and

$$\frac{\partial H(.)}{\partial y} = -\lambda'(t) \quad \frac{\partial^2 H(.)}{\partial u^2} = -2$$

$$\text{Setting } \frac{\partial H}{\partial u} = 0 \Rightarrow -2u + \lambda = 0 \Rightarrow u(t) = \frac{1}{2} \lambda(t)$$

$$\frac{\partial H}{\partial y} = \lambda \text{ and } \frac{\partial^2 H}{\partial y^2} = 0 \quad \text{Similarly } \frac{\partial^2 H(.)}{\partial u \partial y} = \frac{\partial^2 H(.)}{\partial y \partial u} = 0$$

Forming the hessian matrix,  $A = \begin{bmatrix} \frac{\partial^2 H(.)}{\partial y^2} & \frac{\partial^2 H(.)}{\partial y \partial u} \\ \frac{\partial^2 H(.)}{\partial u \partial y} & \frac{\partial^2 H(.)}{\partial u^2} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}$

$\det A = -2 \leq 0$  hence A is

negative definite consequently  $u(t) = \frac{1}{2} \lambda(t)$  ... (1)

maximize the objective function.

From costate equation  $\frac{\partial H(.)}{\partial y} = -\lambda'(t) = \lambda$

We get general solution  $\lambda(t) = k e^{-t}$  ... (2)

at  $t = 0$   $\lambda(0) = k$  hence  $\lambda(t) = \lambda(0) e^{-t}$

using eq (2) in (1)  $u(t) = \frac{1}{2} k e^{-t}$  ... (3)

Using (2), (3) in state dynamic equation  $y' = y + \frac{1}{2} k e^{-t}$

$\Rightarrow y' - y = \frac{1}{2} k e^{-t}$

This is a first order linear differential equation with a variable coefficient and a variable term of type  $\frac{dy}{dt} + u(t)y = w(t)$  here with

$u(t) = -1$  and  $w(t) = \frac{1}{2} k e^{-t}$  via standard formula

$$y(t) = e^{-\int -1 dt} \left( c + \frac{1}{2} \int k e^{-t} e^{-\int -1 dt} dt \right)$$

$$= e^t \left( c + \int k \frac{1}{2} e^{-2t} dt \right)$$

Hence  $y(t) = e^t \left( c - \frac{1}{4} k e^{-2t} \right)$

$$= c e^t - \frac{1}{4} k e^{-t}$$
 ... (4)

where c is an arbitrary constant.

Using the boundary value condition  $y(0) = 1$   $y(1) = 0$

We get  $c = \frac{1}{1 - e^2}$  and  $k = \frac{4e^2}{1 - e^2}$

Putting it altogether

$$y^*(t) = \frac{1}{1-e^2} e^t - \frac{e^2}{1-e^2} e^{-t} \text{ and } \lambda^*(t) = \frac{4e^2}{1-e^2} e^{-t}$$

$$\text{and } u^*(t) = \frac{2e^2}{1-e^2} e^{-t} \text{ (Ans)}$$

## Check Your Progress 2

1) Rewriting the problem  $\min_{u^*} g_0(u) = \int_0^1 (x^2 + u^2) dt$

s.t.  $x' = u$  given  $x(0) = 1$  that means  $x(1)$  is free

The corresponding Hamiltonian  $H(.) = x^2 + u^2 + \lambda u$

Using necessary condition  $\frac{\partial H}{\partial u} = 0 \Rightarrow 2u + \lambda = 0$

$$\Rightarrow u^* = -\frac{1}{2} \lambda^*(t) \quad \dots (1)$$

Using co-state equation  $\frac{\partial H}{\partial x} = -\lambda'(t)$

$$\Rightarrow \frac{d\lambda}{dt} = -2x(t) \Rightarrow x = -\frac{1}{2} \frac{d\lambda}{dt} \quad \dots (2)$$

Using boundary value condition (ref 23.30) here  $t_f = 1$

$$\left[ \frac{\partial S}{\partial x} - \lambda \right]_{t=t_f} = 0 \Rightarrow \lambda(1) = 0 \quad \dots (3)$$

The state dynamic equation  $x' = -u$

$$\Rightarrow \frac{dx}{dt} = \frac{-\lambda(t)}{2} \quad \dots (4)$$

Differentiating both sides of above equation  $\frac{d^2x}{dt^2} = \frac{1}{2} \frac{d\lambda}{dt} = x$  (using

2)

$$\text{Therefore, we can write } \frac{d^2x}{dt^2} - x = 0 \quad \dots (5)$$

For complementary solution let  $Ae^{rt} = x$

then equation (5) can be written  $Ae^{rt}(r^2 - 1) = 0$

as  $Ae^{rt} = 0$  is trivial solution we set  $r^2 = 1$  then  $r = \pm 1$

Hence the solution of equation (5) can be written

$$x(t) = Ae^t + Be^{-t} \quad \dots (6)$$